

Mathematical Analysis and Computer Simulation of the Respiratory System in the Newborn Infant

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Abstract—A mathematical model of neonatal respiratory control is proposed which can be used to simulate the system under different physiological conditions. The model consists of a continuous plant and a discrete controller. Included in the plant are lungs, body tissue, brain tissue, a cerebrospinal fluid compartment, and central and peripheral receptors. The effect of shunt in the lungs is included in the model and the lung volume and the dead space are time varying. The controller utilizes outputs from peripheral and central receptors to adjust the depth and rate of breathing and the effects of prematurity of peripheral receptors are included in the system. Hering-Breuer type reflexes are embodied in the controller to accomplish respiratory synchronization. The model is examined and its simulation results under test conditions in hypoxia and hypercapnia are presented.

I. INTRODUCTION

DIFFERENT researchers have presented a number of mathematical models to simulate the human respiratory system for adults since 1949. Among these researchers, the names of Gray [9], Horgan [13], Milhorn [18], Grodins [10], Longobardo [16], and Yamamoto [28] easily come to mind. In 1983, Fincham and Tehrani [7] presented a model which included many dynamic aspects of the system and reproduced both the low and high frequency effects of different physiological stimuli under transient and steady state conditions.

In recent years, strong evidence indicative of low activity of arterial chemoreceptors in the newborn period has been presented [1], [5]. The respiratory system shows signs of prematurity in the newborn period and tends to lapse into failure because of fluctuations in the lung function. The new discoveries in neonatal respiratory physiology, the respiratory disorders observed immediately after birth, and the fatalities caused by these disorders have prompted the need for the development of a reliable mathematical model for the respiratory system in the newborn infant. Such a model can be used as a valuable aid in the investigation of respiratory disorders and maturation in the neonatal period.

Revow *et al.* [22] presented a model in 1989 which could successfully simulate the respiratory system of the newborn infant during epochs of quiet sleep. In the system developed for this model, the cerebrospinal fluid compartment was not separated from the brain. The model which was based on earlier descriptions of the system for adult respiratory control

[11], [18], was not examined under different physiological conditions.

The purpose of the present paper is to propose a sufficiently descriptive dynamic model for the neonate's respiratory system which can be used as a simulation aid in the study of behavior and maturation of the system in the newborn period without the need for invasive experiments. An earlier model for human adults by Fincham and Tehrani [7] has been used as the basic structure of this model and further developed to include the characteristics of the respiratory system in the newborn period. In this model, the controller acts discretely while the plant is a continuous system. Lungs, brain, and cerebrospinal fluid compartments are separated from each other, the lung volume varies sinusoidally over the breathing cycle and the effect of shunt in the lungs is included. The respiratory work output is considered and the Hering-Breuer reflexes are represented to achieve respiratory synchronization and to allow for the adjustment of the depth and rate of breathing for every breath. The controller adjusts the level of ventilation, the rate of breathing and the dead space volume on the basis of the chemical states of the blood in the neighborhoods of arterial and central chemoreceptors for every breathing cycle. The effects of prematurity of arterial chemoreceptors in the newborn period are included and tested in the model.

The comprehensive set of model equations was solved by using the "Interactive Simulation Language," (ISL), which is a block oriented simulation software on an IBM digital computer.

II. MODEL DESCRIPTION

Fig. 1 shows the block diagram of the model. The system comprises a continuous plant and a discrete respiratory controller. The plant consists of lungs, body tissue, brain tissue, a cerebrospinal fluid compartment and peripheral and central medullary receptors. A transport time delay is included between lungs and the carotid body receptors.

The respiratory controller receives the information about gas tensions at the sites of arterial and central receptors continuously. The mean values of this information are detected by the controller over the period of every breath and the depth and rate of breathing are adjusted for the next breathing cycle.

A. Plant Description

In order to construct the plant, a significant alveolar-arterial oxygen gradient ($A-a$) O_2 of 20 to 30 mm Hg has been

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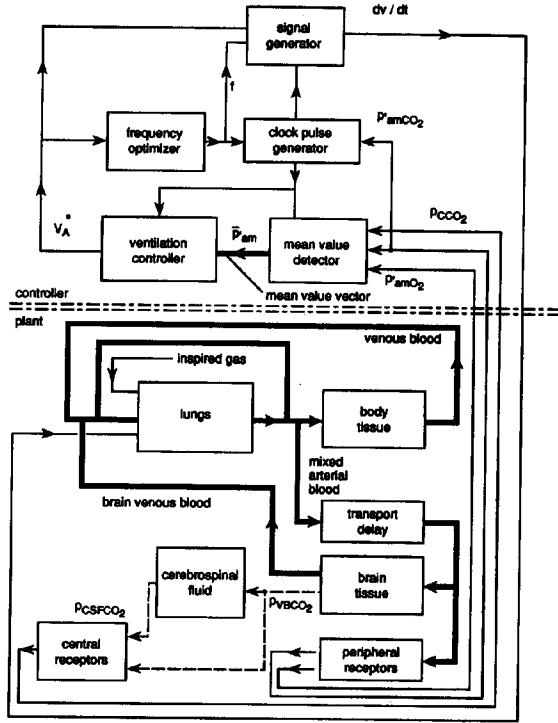


Fig. 1. Model block diagram. — CO₂, O₂ main blood circulation, - - - CSF and central receptor effects of CO₂.

assumed and the effect of the anatomical shunts in the lungs during the neonatal period are included. The plant equations are written by using the symbols listed in the "Appendix."

The mass flow equations for carbon dioxide and oxygen in the alveolar space are:

$$(C_{VTCO_2} - C_{aCO_2})(1 - \alpha)Q_T + (C_{VBCO_2} - C_{aCO_2}) \cdot (1 - \alpha)Q_B = \frac{v}{P_b - 47} \frac{dP_{ACO_2}}{dt} + G1 \quad (1)$$

and

$$(C_{VTO_2} - C_{aO_2})(1 - \alpha)Q_T + (C_{VBO_2} - C_{aO_2}) \cdot (1 - \alpha)Q_B = \frac{v}{P_b - 47} \frac{dP_{AO_2}}{dt} + G2 \quad (2)$$

where

$$\left[\begin{aligned} G1 &= \frac{P_{ACO_2} - P_{ICO_2}}{P_b - 47} \frac{dv}{dt} \\ G2 &= \frac{P_{AO_2} - P_{IO_2}}{P_b - 47} \frac{dv}{dt} \end{aligned} \right]$$

for

$$\frac{dv}{dt} \geq 0 \quad (\text{for inspiration})$$

and for expiration:

$$G1 = G2 = 0 \quad \text{for } \frac{dv}{dt} < 0.$$

In these equations, alveolar and arterial partial pressures of CO₂ are equal:

$$P_{ACO_2} = P_{aCO_2} \quad (3)$$

and for oxygen:

$$P_{AO_2} = P_{aO_2} + K \quad (4)$$

where K is the alveolar-arterial oxygen difference.

Assuming homogeneous mixing of venous blood from the anatomical shunts in the lungs and the arterial blood:

$$C_{amCO_2} = (1 - \alpha)C_{aCO_2} + \frac{\alpha Q_T}{Q} C_{VTCO_2} + \frac{\alpha Q_B}{Q} C_{VBCO_2} \quad (5)$$

$$C_{amO_2} = (1 - \alpha)C_{aO_2} + \frac{\alpha Q_T}{Q} C_{VTO_2} + \frac{\alpha Q_B}{Q} C_{VBO_2} \quad (6)$$

For Lumped Body Tissue:

$$C_{VTCO_2}Q_T = C_{amCO_2}Q_T + MR_{TCO_2} - S_T \frac{dC_{TCO_2}}{dt} \quad (7)$$

$$C_{VTO_2}Q_T = C_{amO_2}Q_T - MR_{TO_2} - S_T \frac{dC_{TO_2}}{dt} \quad (8)$$

Transport Delay in the Arteries:

$$P'_{aCO_2} = P_{amCO_2}(t - H) \quad (9)$$

$$P'_{aO_2} = P_{amO_2}(t - H) \quad (10)$$

Brain Tissue:

$$C_{VBCO_2}Q_B = C'_{amCO_2}Q_B + MR_{BCO_2} - S_B \frac{dC_{BCO_2}}{dt} \quad (11)$$

$$C_{VBO_2}Q_B = C'_{amO_2}Q_B - MR_{BO_2} - S_B \frac{dC_{BO_2}}{dt} \quad (12)$$

Central Medullary Receptors: The partial pressure of carbon dioxide at the site of central receptors P_{CCO_2} is taken to be the CO₂ pressure at a depth, d , below the surface of the medulla and is a function of the partial pressures of carbon dioxide in the cerebrospinal fluid, P_{CSFCO_2} , CO₂ tension in the brain venous blood, P_{VBCO_2} , and the cerebral blood flow Q_B . The equation describing the central medullary receptor is [19]:

$$P_{CCO_2} = P_{VBCO_2} + [P_{CSFCO_2} - P_{VBCO_2}] \cdot \exp[-d(Q_B K_1)^{0.5}]. \quad (13)$$

Cerebrospinal Fluid Compartment: The diffusion equation for CO₂ across the blood-CSF barrier is [3]:

$$\frac{dP_{CSFCO_2}}{dt} = \frac{1}{K_2} (P_{VBCO_2} - P_{CSFCO_2}) \quad (14)$$

Blood Gas Relationships: A linear CO_2 dissociation curve is assumed for carbon dioxide [23]:

$$C_{\text{CO}_2} = K_3 P_{\text{CO}_2} \quad (15)$$

and for oxygen [27]:

$$C_{\text{O}_2} = K_4 (1 - e^{-K_5 P_{\text{O}_2}})^2. \quad (16)$$

Although cardiac output and cerebral blood flow are both functions of arterial pressures of carbon dioxide and oxygen [8], but due to the lack of sufficient experimental data for infants, these variables are considered to remain constant in the model.

B. Description of the Controller

The respiratory controller is represented by a discrete system in the model. The controller consists of a mean value detector, a ventilation controller, a frequency optimizer, a driving signal generator, and a clock pulse generator which synchronizes the functions of the other blocks by producing a pulse at the end of every breathing cycle. The synchronizing effect of the clock pulse generator in this model simulates the Hering-Breuer reflexes in the respiratory control system.

The information about blood gas concentrations at the sites of the arterial and central receptors is supplied to the controller continuously. The mean value detector calculates the average values of this data over the interval of every breath and provides the calculated data to the ventilation controller.

Ventilation Controller: The following equation which is derived in [7] has been used to determine the alveolar ventilation:

$$\frac{V_A}{V_{A(\text{rest})}} = 0.2025 \bar{P}'_{am\text{CO}_2} + 0.2332 \bar{P}_{CC\text{CO}_2} + F1 - \beta. \quad (17)$$

In this equation, β is a constant and $F1$ is the response of the peripheral receptors to disturbances in oxygen balance. The above equation has been modified to be used in the neonatal respiratory controller by considering $F1$ to be a function of the maturity of the peripheral chemoreceptors in the newborn infant. $F1$ is defined as

$$F1 = 4.72 \times 10^{-9} \mu (104 - P'_{am\text{O}_2})^{4.9}$$

$$\text{for } P'_{am\text{O}_2} < 104 \text{ mm Hg}$$

$$\text{and } F1 = 0 \text{ for } P'_{am\text{O}_2} \geq 104 \text{ mm Hg}$$

$$\text{where } 0 < \mu \leq 1. \quad (18)$$

In this equation, μ is a parameter which depends on maturity and hypoxic sensitivity of the peripheral chemoreceptors in the newborn infant. Using (17) and (18), the required level of alveolar ventilation is determined for every breath. If due to acute hypoxic conditions, the partial pressure of carbon dioxide in the mixed arterial blood decreases below a threshold

value of 30 mm Hg [20] or the required calculated alveolar ventilation turns out to be negative under such conditions, ventilation is reduced to zero and the calculations are repeated until (17) and (18) produce a positive value for ventilation and blood P_{CO_2} increases above the threshold level.

Frequency Optimizer: The frequency of breathing is determined by minimizing the respiratory work output. The equation for the optimum frequency of breathing on the basis of minimum work criterion was found by Otis *et al.* [21]. By considering the lungs' elastance and airway resistance and ignoring the turbulence forces, the frequency of breathing is given by

$$f = \frac{-\gamma V_D + \sqrt{(\gamma V_D)^2 + 4\gamma\lambda\pi^2 V_D V_A}}{2\pi^2 \lambda V_D} \quad (19)$$

In this equation, V_D is the dead space volume which is found by

$$V_D = 0.1698 V_A + 2.28 \times 10^{-3}. \quad (20)$$

Equation (20) is an empirical relationship which is based on experimental data for the newborn infant [14], [25].

Driving Signal Generator: Air flow into the lungs is represented by a sinusoidal waveform in the model. The driving signal generator in the respiratory controller produces this waveform and adjusts its amplitude and frequency for every breath according to the following equation:

$$\frac{dv}{dt} = \pi V_A \sin 2\pi ft. \quad (21)$$

The expired volume is a function of alveolar ventilation, frequency of breathing and the dead space volume and is given by

$$V_E = V_A + f1.V_D. \quad (22)$$

III. RESULTS AND DISCUSSION

The simulation model was examined under different physiological conditions including rest, hypoxia and hypercapnia. The shunt ratio in the lungs was assumed to be 0.1 and a significant alveolar-arterial oxygen difference of 20 mm Hg was used in all the simulation experiments. Values of the other model parameters are given in the Appendix which correspond to the respiratory system of a one week old infant. Table I shows the steady state values of a selection of the variables of the system after 8 min excitation. Under hypoxic conditions, the steady state level of partial pressure of oxygen in the mixed arterial blood, $P'_{am\text{O}_2}$, increases by increasing the control parameter μ , which is an indicator of functionality of peripheral chemoreceptors in response to oxygen disturbances. Alveolar ventilation also increases with μ and consequently the steady state level of partial pressure of carbon dioxide in the mixed arterial blood, $P'_{am\text{CO}_2}$, decreases. The system shows significant sensitivity to control parameter μ under acute hypoxic conditions. The simulation results in hypercapnia do not show much sensitivity to μ as expected. Under hypercapnic conditions, $P'_{am\text{CO}_2}$ increases and causes a significant percentage increase in alveolar ventilation. As a result of the intervention of the controller and high ventilation,

TABLE I

THE STEADY-STATE SIMULATION RESULTS OF THE MODEL AT REST, HYPOXIA, AND HYPERCAPNIA AFTER 8 min. OF STIMULATION. V_A AND V_E ARE IN l/min., V_D IS IN l, AND f_1 IS THE BREATHING FREQUENCY IN BREATH/MIN. SHUNT RATIO (α) IS 0.1 AND ALVEOLAR-ARTERIAL OXYGEN DIFFERENCE (K) IS 20 mm Hg. THE VALUES OF THE OTHER PARAMETERS ARE GIVEN IN THE APPENDIX

Test Condition		Simulation Results							
F_{IO_2}	F_{ICO_2}	μ	P_{amO_2}	P_{amCO_2}	P_{CCO_2}	V_A	V_D	V_E	f_1
Resting Conditions									
0.21	0	1.0	73	39.4	41.8	0.491	3.70×10^{-3}	0.6	30.0
Hypoxia									
0.15	0	1.0	48.4	35.5	39.1	0.623	4.04×10^{-3}	0.754	32.5
0.15	0	0.8	47.9	36.5	39.4	0.614	4.02×10^{-3}	0.743	32.3
0.15	0	0.5	46.1	36.8	40.1	0.481	3.64×10^{-3}	0.589	29.8
0.12	0	1.0	38.1	30.3	35.4	0.806	4.56×10^{-3}	0.965	35.0
0.12	0	0.8	36.5	31.3	36.2	0.778	4.48×10^{-3}	0.933	34.7
0.12	0	0.5	34.2	33	37.6	0.649	4.12×10^{-3}	0.784	32.88
Hypercapnia									
0.21	0.03	1.0	98.8	40.8	43.4	0.848	4.68×10^{-3}	1.014	35.5
0.21	0.03	0.8	98.7	40.8	43.4	0.848	4.68×10^{-3}	1.014	35.5
0.21	0.03	0.5	98.6	40.8	43.4	0.849	4.68×10^{-3}	1.015	35.5
0.21	0.05	1.0	106.6	44.8	46.2	1.869	7.57×10^{-3}	2.187	42.1
0.21	0.05	0.8	106.7	44.5	46.1	1.826	7.45×10^{-3}	2.138	41.9
0.21	0.05	0.5	106.7	44.3	46.2	1.804	7.38×10^{-3}	2.114	41.8

P'_{amO_2} rises and approaches the threshold level of 104 mm Hg. Since the arterial chemoreceptors do not respond to oxygen disturbances above this threshold level [see (18)], the steady-state simulation results are almost unaffected by changes in μ under different hypercapnic conditions. The results of Table I show large percentage changes in ventilation and breathing frequency in acute hypercapnia and smaller changes in these variables under hypoxic conditions. These simulation results are in general agreement with experimental observations [2], [14], [17].

Examples of transient responses of the model to different stimuli are presented in Figs. 2–5. In Fig. 2, the sinusoidal airflow signal at rest has been shown. In this experiment, the value of the control parameter μ is 1.0 (corresponding to fully matured peripheral chemoreflexes). The airflow signal oscillates between 0.035 and -0.035 L/s with an alveolar ventilation of about 10 mL/s at the early stages of simulation. The steady oscillations indicate regional stability of the system under resting conditions.

The transient responses of the model in 15% oxygen breathing are shown in Fig. 3. Mean values of mixed arterial pressures of oxygen and carbon dioxide are plotted against time for 500 s. The control parameter μ is assumed to be 0.5. During the first 40 seconds of hypoxia, there is a rapid fall in P'_{amO_2} which is the result of the reduction in inspired oxygen fraction, F_{IO_2} . The decline of P'_{amO_2} continues in the transition region at a slower rate due to the effect of venous tension of oxygen which in turn is affected by the arterial oxygen pressure. The secondary effect of oxygen appears less quickly in the transition period due to the storage volumes of the brain and the rest of the body tissues. The reduction in arterial oxygen pressure causes the alveolar ventilation to increase in the transition period. Finally,

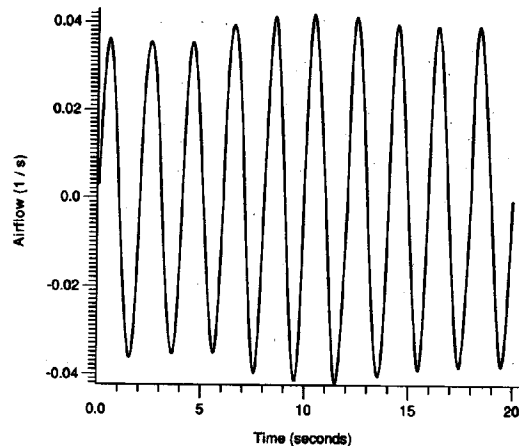


Fig. 2. Simulated airflow pattern at rest, $\alpha = 0.1$, $K = 20$ mm Hg and $\mu = 1.0$.

by physiological intervention of the respiratory controller and increase in ventilation, P'_{amO_2} settles around 46 mm Hg in about 8 min. The rise in alveolar ventilation in hypoxia brings about reduction in the pressure of carbon dioxide in the mixed arterial blood, P'_{amCO_2} . The fall of this variable acts as a powerful antagonist against the lack of oxygen in control of ventilation. P'_{amCO_2} settles around 36.8 mm Hg under steady state conditions in this experiment.

Fig. 4 shows the transient responses of the model to 12% oxygen breathing for $\mu = 0.8$ and $\mu = 0.5$. Comparing the two sets of results, both show a rapid and almost identical fall in P'_{amO_2} during the first 40 s of hypoxia. For $\mu = 0.8$, P'_{amO_2} and P'_{amCO_2} settle around 36.5 mm Hg and 31.3

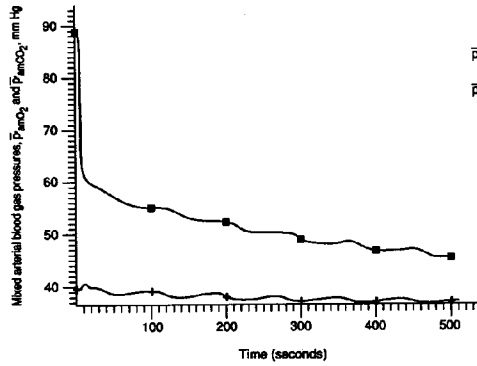


Fig. 3. Response of the model to 15% oxygen breathing. $\alpha = 0.1$, $K = 20$ mm Hg, and $\mu = 0.5$.

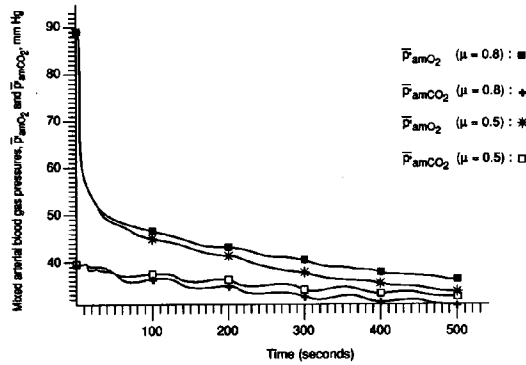


Fig. 4. Comparison of the responses of the model to 12% oxygen breathing for different values of μ . In both experiments $\alpha = 0.1$ and $K = 20$ mm Hg. The values of the control parameter μ in the two experiments are 0.8 and 0.5.

mm Hg, respectively, in about 8 min. For $\mu = 0.5$, P'_{amO_2} and P'_{amCO_2} decrease at a different rate during the transition period and settle around 34.2 and 33 mm Hg, respectively. The results of Fig. 4 show the sensitivity of the transient response of the system to the control parameter, μ , under hypoxic conditions. With a higher value of μ , the activity of peripheral chemoreceptors in hypoxia increases and alveolar ventilation rises to a higher level in response to hypoxic disturbances. The induced hyperventilation brings about higher levels of P_{O_2} and lower levels of P_{CO_2} in the blood as reflected in the results of Fig. 4. The simulation results of Figs. 3 and 4 in hypoxia are in close agreement with experimental data reported in the literature [2], [17].

The responses of the model in hypercapnia and recovery are illustrated in Fig. 5 for $\mu = 0.8$ and $\mu = 0.5$. In the simulation experiments of Fig. 5, the concentration of carbon dioxide in the inspired gas is increased to 5% during the first 500 seconds of excitation. P'_{amCO_2} increases rapidly by 4 mm Hg during the first 200 s of stimulation in both experiments due to the increase in inspired carbon dioxide fraction, F_{ICO_2} . There is a further and slower increase in the value of this variable which reflects the effects of the long diffusion time constant of 320

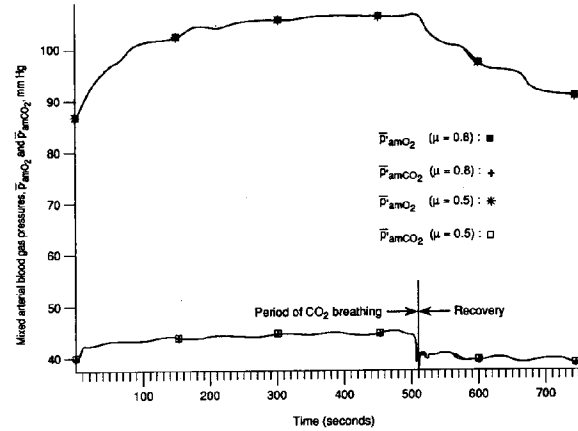


Fig. 5. Comparison of the responses of the model to 5% carbon dioxide breathing and recovery for different values of μ . $\alpha = 0.1$, $K = 20$ mm Hg and the parameter has the values of 0.8 and 0.5 in the simulation experiments.

s associated with the cerebrospinal fluid compartment and the storage volumes of carbon dioxide in the brain and the body tissues. The increase in P'_{amCO_2} causes alveolar ventilation to rise and P'_{amCO_2} reaches a steady state level of 44.5 mm Hg in less than 400 s by the intervention of the controller. P'_{amO_2} also rises rapidly during the first 200 s of these experiments, mainly due to induced hyperventilation. This variable settles around 106.7 mm Hg under steady-state conditions. After the first 500 s of simulation, the infant is exposed to fresh air. During recovery, P'_{amO_2} falls and approaches its normal level at rest in about 500 s as ventilation returns to normal. P'_{amCO_2} falls sharply at the beginning of recovery and after a slight undershoot settles around its steady state value at rest in about 200 s. As can be seen in the results of Fig. 5, the responses of the model are very similar and almost identical for different values of μ and the transient performance of the system in hypercapnia does not show any considerable sensitivity to the value of this parameter. The steady state values of P'_{amO_2} and P'_{amCO_2} are only slightly different in these two experiments and the durations of the transition periods in hypercapnia and recovery are not affected by changing the value of μ .

In Figs. 3–5, there can also be seen slight and steady oscillations in the system variables under steady-state conditions. These steady oscillations are due to the dynamic nature of the model and the fact that breathing is of a cyclical nature as opposed to that of a constant through flow-type in this system.

IV. CONCLUSIONS

A mathematical model has been presented which can be used to simulate the neonatal respiratory control system and to study the effects of prematurity of peripheral chemoreceptors on the respiratory function in the newborn period. In this model, a relatively detailed discrete controller has been used which adjusts the level of ventilation and optimizes the frequency of breathing to minimize the respiratory work

output for every breath. The controller uses inputs from central and peripheral chemoreceptors. Reliable evidence presented by different researchers suggest that hypoxic sensitivity of the peripheral chemoreceptors is premature at birth and is developed to reach its normal functionality in the newborn period [1], [4], [5]. On the basis of these findings, the effects of prematurity of peripheral chemoreceptors are included in the model. Also, Hering-Breuer reflexes are embodied in the controller and a sinusoidal drive signal is generated and supplied to the lungs at the end of every breath. The plant which is a continuous system consists of lungs, brain, body tissues, a cerebrospinal fluid compartment, and peripheral and central chemoreceptors. The effects of the anatomical shunts in the lungs and a varying dead space in the newborn period are included in the system. In the derivation of the system equations, it is assumed that the gas tensions in the tissue are equal to those in the venous blood. Also, homogeneous mixing of venous blood from the brain and body tissues prior to the lungs and similarly homogeneous mixing of venous blood from the anatomical shunts in the lungs and the arterial blood have been assumed. Cardiac output and cerebral blood flow which are known to be functions of arterial pressures of oxygen and carbon dioxide [8] are assumed to remain constant in the model due to lack of sufficient experimental data for neonates.

The performance of the model was evaluated using a wide range of stimuli and the transient and steady state behavior of the model were examined. Instability was never observed under any test condition, indicating general stability of the system. The simulation results were compared with clinical data whenever possible and they were found to be in general agreement with the available experimental values.

In brief, a mathematical model of the respiratory control system in adults has been modified, parameterized and further developed to include the characteristics of the system in the newborn period. Due to relative sophistication of the model and its isomorphism with the real system, it shows the flexibility and the potential to describe the neonatal respiratory control system under a wide range of test conditions and to be used as an effective simulation aid in the analysis and study of respiratory disorders and maturation of the system during the neonatal period.

APPENDIX

GLOSSARY OF SYMBOLS AND VARIABLES

The main symbols specify the variable or parameter, and subscripts denote the location in the model or a specific chemical.

Model Symbols and Parameters		
Symbol/ Variable	Definition	Value/units
C	Volume concentration of gas in the blood	$[l(\text{STPD})/l]$
d	Depth of central receptor below surface of the medulla	$[15 \times 10^{-3} \text{ cm}]$

F	Fractional composition of gas	
f	Frequency of breathing	(breath/s)
f_1	Frequency of breathing	(breath/min)
K	Alveolar-arterial oxygen difference	(mm Hg)
K_1	Central receptor constant	$(346 \times 10^3 \text{ s cm}^{-2})^{-1}$
K_2	Carbon dioxide diffusion time constant across the blood-CSF barrier	(320 s)
K_3	Blood gas dissociation constant for carbon dioxide	$[0.016 (\text{mm Hg})^{-1}]$
K_4	Blood gas dissociation constant for oxygen	(0.2)
K_5	Blood gas dissociation constant for oxygen	$[0.046 (\text{mm Hg})^{-1}]$
β	Ventilation controller constant	(17.1756)
γ	Respiratory system elastance	$[91 \text{ cm H}_2\text{O}/l]$
λ	Airway resistance in the lungs	$[60 \text{ cm H}_2\text{O}/l(\text{BTPS})/\text{s}]$
MR_B	Metabolic rate in brain	(l/s)
	basal rates CO_2	(2.6154×10^{-4})
	O_2	(2.78697×10^{-4})
MR_T	Metabolic rate in body tissue	(l/s)
	basal rates CO_2	(1.625×10^{-4})
	O_2	(1.902×10^{-4})
P	Partial pressure or gas pressure	(mm Hg)
Q	Blood flow rate	(l/s)
	basal values Q	(13.13×10^{-3})
	Q_T (tissue)	(9.8×10^{-3})
	Q_B (brain)	(3.33×10^{-3})
S	Equivalent gas storage space	(l)
	S_T (tissue)	(0.9)
	S_B (brain)	(0.067)
H	Transport delay (arterial blood)	(4s)
t	time	
V	Volume	(l)
V_A	Alveolar ventilation	(l/s)
V_A	Alveolar ventilation	(l/min)
V_E	External ventilation	(l/min)
v	Volume of the lungs	(l)
α	Shunt ratio in the lungs	
μ	Parameter of the respiratory controller	

Subscripts

<i>A</i>	Alveolar gas
<i>a</i>	Arterial blood
<i>am</i>	Mixed arterial blood
<i>B</i>	Brain
<i>b</i>	Barometric condition
<i>C</i>	A point in the medulla (neighborhood of the central receptors)
CO_2	Carbon dioxide
CSF	Cerebrospinal fluid
<i>D</i>	Dead space
<i>E</i>	External ventilation
<i>I</i>	Inspired gas
O_2	Oxygen
<i>T</i>	Body tissue (excluding brain)
<i>V</i>	Venous blood
<i>VB</i>	Brain venous blood
<i>VT</i>	Tissue venous blood

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